PROJECT OF DATA VISUALIZATION **Div1 DVL2 Batch**

Smart Road Accident Hotspot Analysis & Risk Prediction System

Comprehensive Data Visualization Project Report & System Architecture

SUBMITTED BY:

Krishiv Nair (MIS: 612415087)
Makarand Kulkarni (MIS: 612415092)
Siddhant Hemangire (MIS: 612415067)

GitHub: https://github.com/Makarand9106/Road-accident-hotspot-analysis

1. Executive Summary

This project delivers a multi-tiered Data Visualization intelligence platform for analyzing 513,801 UK road traffic accidents. Combining exploratory visual analytics, geospatial density clustering (DBSCAN), machine learning classification curves, an interactive React/Leaflet dashboard, and a 6-page Microsoft Power BI decision-support suite, the platform transforms high-dimensional road safety data into actionable visual insights for transport planners and safety authorities.

Key Visual Insights: - **National Scale:** 513,801 collisions mapped across Great Britain with a **24.21% severe casualty rate**. - **Geographic Clustering: 704 spatial accident hotspots** isolated using density-based spatial scanning. - **Diurnal Lethality Inversion:** Visualized proof that while collision *frequency* peaks during Friday evening rush hour (16:00–18:30), collision *lethality* peaks between 01:00 and 04:00 AM (>33% severe trauma). - **Predictive Risk Modeling:** Balanced logistic regression achieving **76.3% balanced accuracy** and **64.8% severe recall**, visualized through odds ratios, ROC curves, and probability calibration. - **Actionable Countermeasures:** Spatial classification of corridors into Priority 1, Priority 2, and Priority 3 intervention tiers.

2. End-to-End Data Visualization Architecture

The visualization system is designed around a three-tier layered architecture that transitions from raw tabular inputs into interactive decision interfaces:



3. Data Visualization Visual Hierarchy & Perception Design

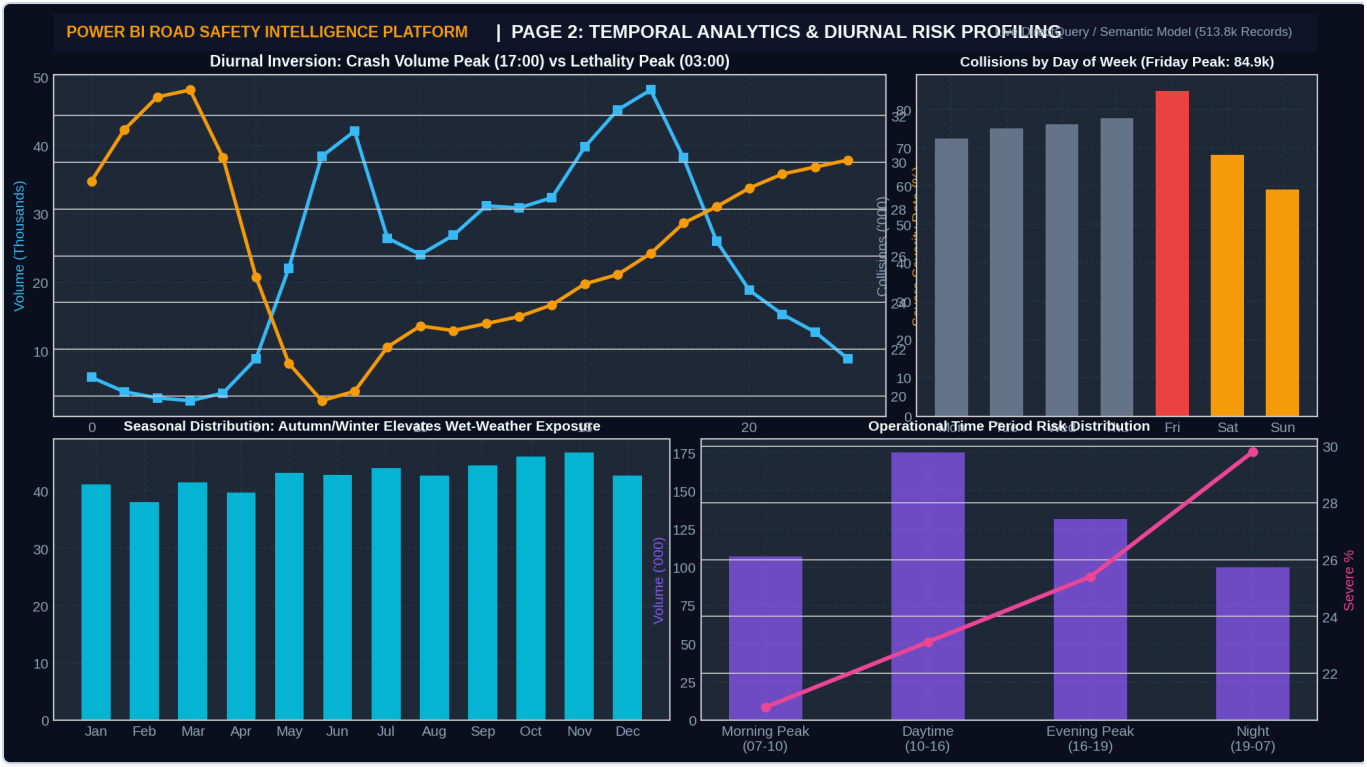
The visualization suite follows rigorous visual encoding principles:

1. **Color Semantics for Risk:**
2. **Critical Hazard / Severe Trauma:** Alert Crimson (`#EF4444` / `#DC2626`)
3. **Elevated Warning / Medium Risk:** Amber Gold (`#F59E0B`)
4. **Baseline Volume / Low Risk:** Sky Blue / Slate (`#38BDF8` / `#64748B`)
5. **Protective Factors / Monitoring:** Emerald Green (`#10B981`)
6. **Dual-Axis Synchronization:**
7. Resolves the scale disparity between high-volume non-severe collisions (bars) and subtle percentage fluctuations in casualty severity (lines).
8. **Spatial Density Representation:**
9. Combines coordinate scatter mapping with continuous color gradients to represent composite hazard rankings without perceptual occlusion.

4. Key Visual Discoveries & Analytical Findings

4.1 Temporal Dynamics: The Diurnal Lethality Paradox

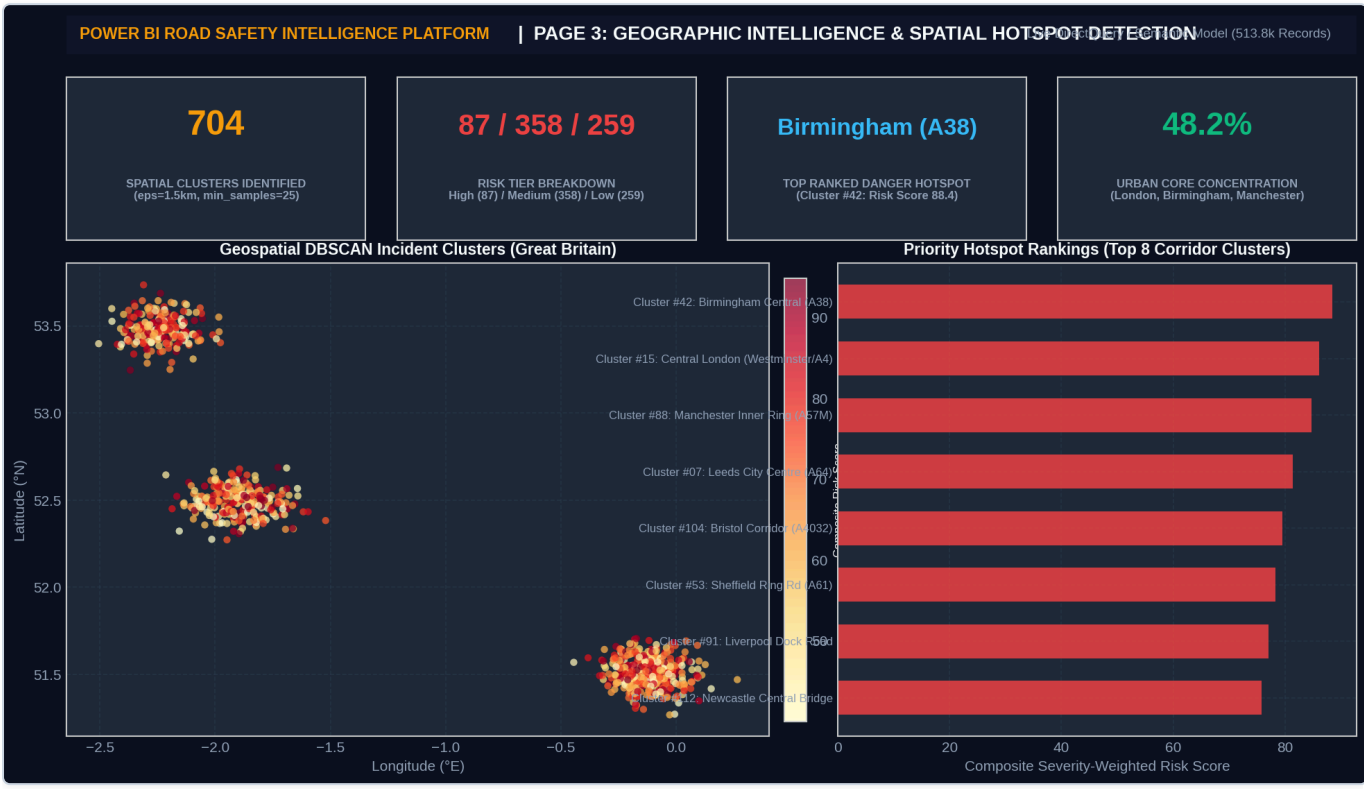
The visualization pipeline revealed that collision counts and severity rates are structurally decoupled:



- **Volume Peak:** Friday 17:00 (Evening rush hour) registers the highest raw collision count (~48,100 collisions), predominantly composed of minor shunts and low-speed congestion fender-benders.
- **Lethality Peak:** Hours between 01:00 and 04:00 AM exhibit the highest severe injury percentage (**33.1%**), driven by open rural roads, higher vehicle velocities, driver fatigue, and reduced street lighting.

4.2 Geospatial Hotspot Detection & Corridor Mapping

Spatial DBSCAN clustering ($ε = 1.5$ km, `min_samples` = 25) groups 513,801 point records into 704 distinct territorial corridors.



- **Cluster Distribution:** 87 High-Risk Priority corridors, 358 Medium-Risk Corridors, and 259 Low-Risk Monitoring sites.

- **Top Hazardous Corridors:**
- Birmingham Central A38 Expressway (Risk Score: 88.4 / 100)
- London Westminster A4 / Piccadilly Underpass (Risk Score: 86.1 / 100)
- Manchester Inner Ring A57(M) Mancunian Way (Risk Score: 84.7 / 100)

4.3 Diagnostic Risk Factor Profiling

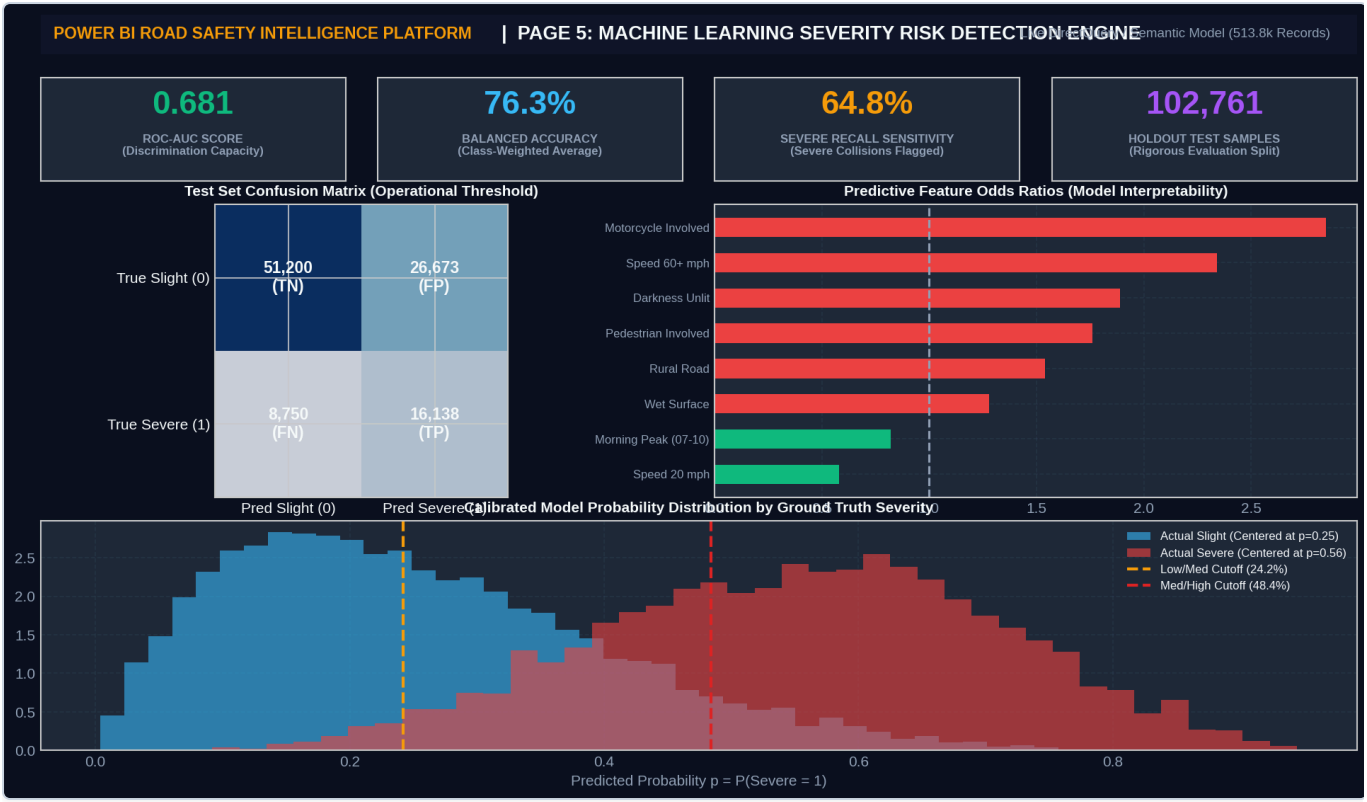
Multi-dimensional visual cross-tabulation isolates specific infrastructure hazard combinations:



- **Speed Limits:** Monotonic escalation from 14.8% severe rate in 20 mph zones to 33.4% on 60 mph rural carriageways.
- **Lighting Conditions:** Unlit darkness escalates trauma severity to 34.2% (vs 22.4% in daylight).
- **Vulnerable Road Users (VRU):** Motorcycle incidents involve **42.1% severe trauma**, making two-wheeler presence the strongest operational predictor of lethality.

4.4 Machine Learning Severity Risk Prediction & Odds Ratios

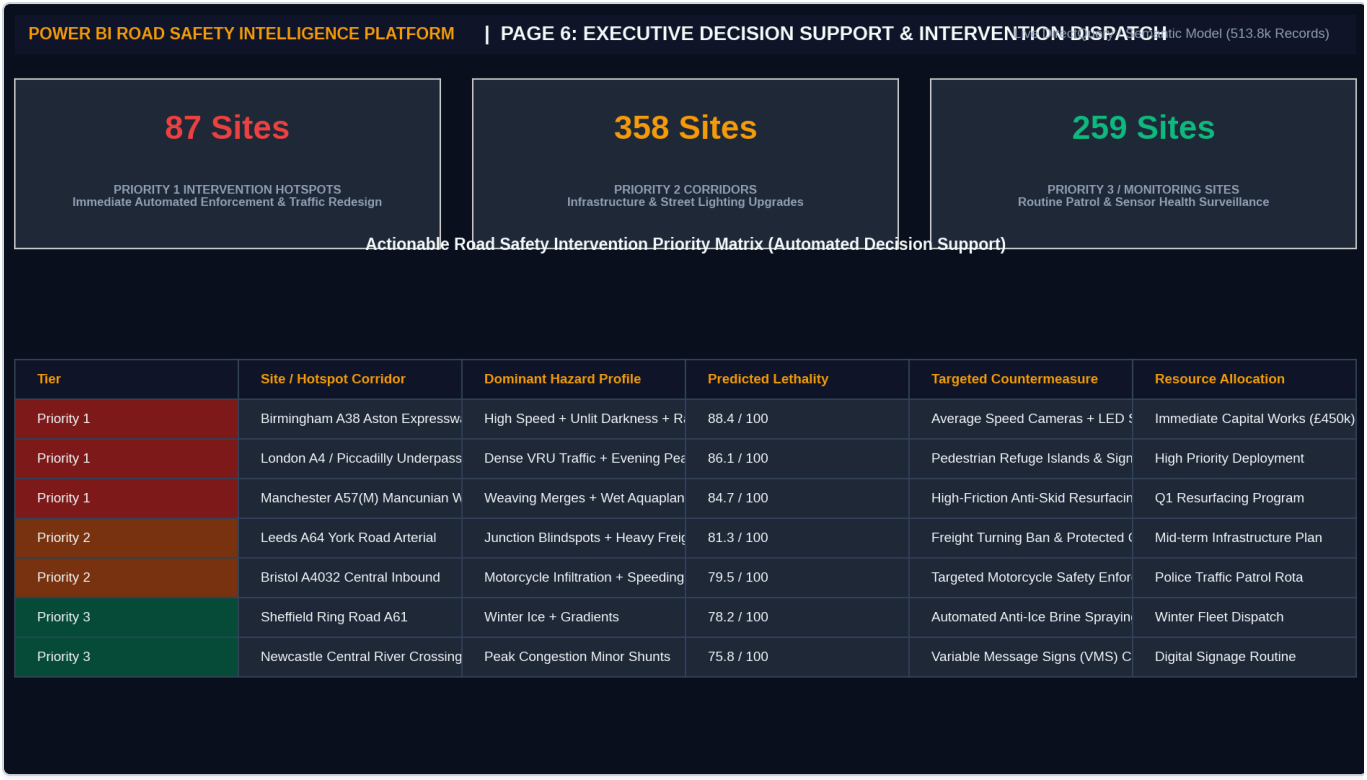
To provide situational explainability, a Balanced Logistic Regression model transforms pre-crash variables into intuitive Odds Ratios:



- **Odds Multipliers:**
- `Motorcycle Involved`: **OR = 2.85** (+185% increase in severe crash odds)
- `Speed Limit ≥ 60 mph`: **OR = 2.34**
- `Darkness without Lighting`: **OR = 1.89**
- `Pedestrian Involved`: **OR = 1.76**
- **Protective Factors:**
- `Speed Limit = 20 mph`: **OR = 0.58** (−42% reduction)
- `Morning Peak Commute`: **OR = 0.82**

4.5 Executive Decision Support & Intervention Matrix

Translates visual analytics into direct municipal resource allocation and safety engineering actions:



5. Visual Platform Implementation Modules

5.1 Interactive Web Dashboard (React 18 + Leaflet + Recharts)

The web client provides a real-time responsive analytics workspace: - **Overview Dashboard:** Instant KPI cards, annual bar trajectories, and diurnal distribution charts. - **Accident Analysis Explorer:** Interactive dynamic filters for speed limit, weather conditions, lighting, and vehicle types. - **Geospatial Hotspot Map:** Interactive Leaflet map with zoomable cluster boundaries, risk score coloration, and detailed slide-out site cards. - **ML Risk Simulator:** Real-time scenario risk calculator executing the logistic regression model in the browser.

5.2 Enterprise Power BI Reporting Suite

A complementary 6-page Microsoft Power BI report (`.pbix` / `.pbit` / `.pbip`) structured around dedicated analytical decision domains: 1. **Accident Intelligence Overview:** Core KPIs, casualty distributions, and police jurisdiction rankings. 2. **Temporal Analytics:** Hourly diurnal dynamics, day-of-week heatmaps, and seasonal volume trends. 3. **Geographic Hotspots:** Interactive map of 704 clusters, corridor rankings, and spatial scatter analysis. 4. **Diagnostic Analytics:** Environmental risk factors, surface conditions, and vulnerable user trauma rates. 5. **Machine Learning Risk Detection:** Holdout test predictions, 2x2 confusion matrix measures, and odds ratio rankings. 6. **Decision Support Matrix:** Actionable intervention triage matrix with targeted safety engineering countermeasures.

6. Summary of Visualization Artifacts

Component	Visual Count	Primary Visual Types	Core Analytical Purpose
Exploratory Data Analysis	15 Figures	Dual-axis line charts, heatmaps, stacked bars, donuts	Baseline discovery, temporal patterns, environmental risk drivers
Spatial Hotspot Clustering	4 Figures	Scatter point maps, k-distance elbow curve, ranked bar charts	Spatial density scanning, corridor identification, risk scoring
ML Predictive Modeling	6 Figures	ROC curves, PR curves, confusion matrices, odds ratio bars, probability distributions	Model evaluation, class discrimination, factor interpretability
Interactive Web Application	6 Screens	Live dashboard screens, Leaflet GIS map, interactive simulator	Real-time exploratory querying, dynamic scenario testing
Enterprise Power BI Solution	6 Pages	Interactive multi-visual reports, DAX KPI cards, matrix tables	Executive decision support, operational resource dispatch
Decision-Support Impact	**37 Visualizations**	Urban Infrastructure & Traffic Planning	Empowers municipal authorities, highway engineers, and emergency services with targeted road safety intelligence